

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

SECOND PARTIAL EXAMINATION - A.Y. 2023/2024

June 3, 2024

Design of a Controller

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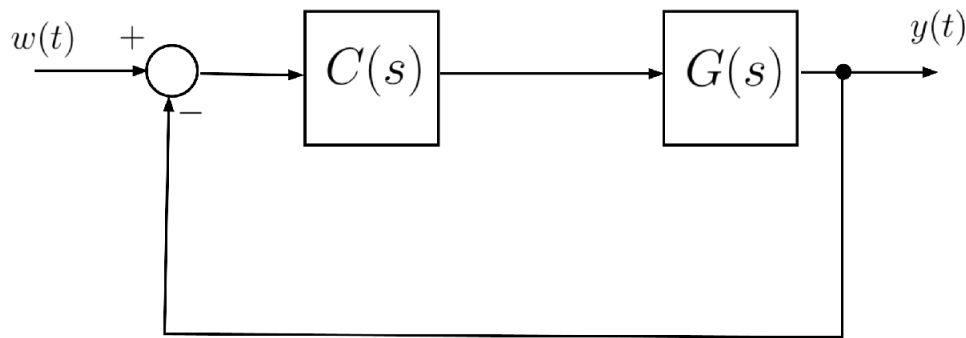
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Description of the Feedback Control System

Consider the feedback control system described by the block-scheme shown in the following figure



where $G(s)$ is given by

$$G(s) = \frac{4.815}{s(1 + 0.1s)(1 + 10s)}$$

Question

Q1: Design a controller $C(s)$ such that all following requirements are simultaneously met:

1. $|e(\infty)| \leq 0.2$ for $w(t) = t \cdot \mathbf{1}(t)$, where $\mathbf{1}(t)$ denotes the unit step function;
2. $t_{s1\%} \leq 2s$, where $t_{s1\%}$ is the settling time at 1 % ;
3. $\Delta\% \leq 20\%$, where $\Delta\%$ is the maximum percentage overshoot, defined by

$$\Delta\% = 100 \cdot \frac{y_p - y(\infty)}{y(\infty)}$$

where y_p is the peak-value of the output.

Hint: the maximum percentage overshoot $\Delta\%$ depends on the damping factor ξ of the closed-loop dominant poles (refer to Part 7 of the course material) and there is also an approximate relationship between the phase margin φ_m and the damping factor of the closed-loop dominant poles (refer to Part 9 of the course material).

A1: Answer to Question Q1

Preliminary Analysis of the Requirements

Req. (Q1.1): According to the requirement, the open-loop system $L(s) = C(s) \cdot G(s)$ has to be a "type $g = 1$ " system [i.e., null steady-state error $e(\infty)$ when the reference input $w(t)$ is a unit step signal ($w(t) = \mathbf{1}(t)$) and the magnitude of the steady-state error must be bounded when the reference input is an unit-slope ramp $w(t) = t \cdot \mathbf{1}(t)$].

Thus, as preliminary controller, let's apply a simple proportional controller $C_1(s) = \mu_1 > 0$ and evaluate the steady-state error when the reference input is $w(t) = t \cdot \mathbf{1}(t)$

$$\text{Hyp. : closed-loop system as. stable} \implies e(\infty) = \lim_{s \rightarrow 0} s \left[S(s) \cdot \frac{1}{s^2} \right], \frac{1}{s^2} = \mathcal{L}(t \cdot \mathbf{1}(t)), \quad S(s) = \frac{1}{1 + L(s)}, \quad L(s) = C_1(s) \cdot G(s)$$

Thus

$$L(s) = \frac{4.815 \mu_1}{s(1 + 0.1s)(1 + 10s)} \implies e(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{s(1 + 0.1s)(1 + 10s)}{s(1 + 0.1s)(1 + 10s) + 4.815 \mu_1} \cdot \frac{1}{s^2} = \frac{1}{4.815 \mu_1}$$

Finally

$$|e(\infty)| \leq 0.2 \implies \left| \frac{1}{4.815 \mu_1} \right| \leq \frac{1}{5} \implies \mu_1 \geq \frac{5000}{4815} \approx 1.04$$

Req. (Q1.2): According to (Part 9 - Slide 44) of the course material

$$\begin{cases} \varphi_m > 75^\circ \implies t_{s\,1\%} \approx \frac{4.6}{\omega_c} \\ \varphi_m < 75^\circ \implies t_{s\,1\%} \approx \frac{4.6}{\xi \omega_c}, \xi \approx \frac{\varphi_m}{100} \end{cases}$$

Thus

$$t_{s\,1\%} \leq 2\text{ s} \implies \begin{cases} \varphi_m > 75^\circ \implies \frac{4.6}{\omega_c} \leq 2 \implies \omega_c \geq \frac{4.6}{2} = 2.3 \\ \varphi_m < 75^\circ \implies \frac{\varphi_m}{100} \omega_c \geq \frac{4.6}{2} = 2.3 \implies \varphi_m \cdot \omega_c \geq 230 \end{cases}$$

Req. (Q1.3): According to (Part 7 - slide 27) of the course material

$$\Delta\% = 100 \cdot e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}} \implies 100 \cdot e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}} \leq 20 \implies e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}} \leq \frac{1}{5}$$

After some manipulations, you obtain

$$\frac{\xi \pi}{\sqrt{1-\xi^2}} \geq \ln 5 \implies \xi \geq \frac{\ln 5}{\sqrt{\pi^2 + (\ln 5)^2}} \approx 0.456$$

Now, according to (Part 9 - Slide 43) of the course material, it holds $\xi \approx \frac{\varphi_m}{100}$

Thus;

$$\xi \geq 0.456 \implies \varphi_m > 45^\circ$$

Summing Up:

- **Req. (Q1.1):** $\mu_1 \geq \bar{\mu}_1 = 1.04$, where $C(s) = \mu_1$
- **Req. (Q1.2):** $\begin{cases} \varphi_m > 75^\circ \implies \omega_c \geq \frac{4.6}{2} = 2.3 \\ \varphi_m < 75^\circ \implies \varphi_m \cdot \omega_c \geq 230 \end{cases}$
- **Req. (Q1.3):** $\varphi_m > 45^\circ$

Solution 1

Design Strategy

- Check if it is possible to meet all the requirements, by imposing $\varphi_m > 75^\circ$

- If not, then check if feasible to obtain $\varphi_m \approx 50^\circ$ or (better choice) $\varphi_m \approx 60^\circ$, by canceling the "slow" pole, while meeting the other two requirements.

Design Procedure

```
clear

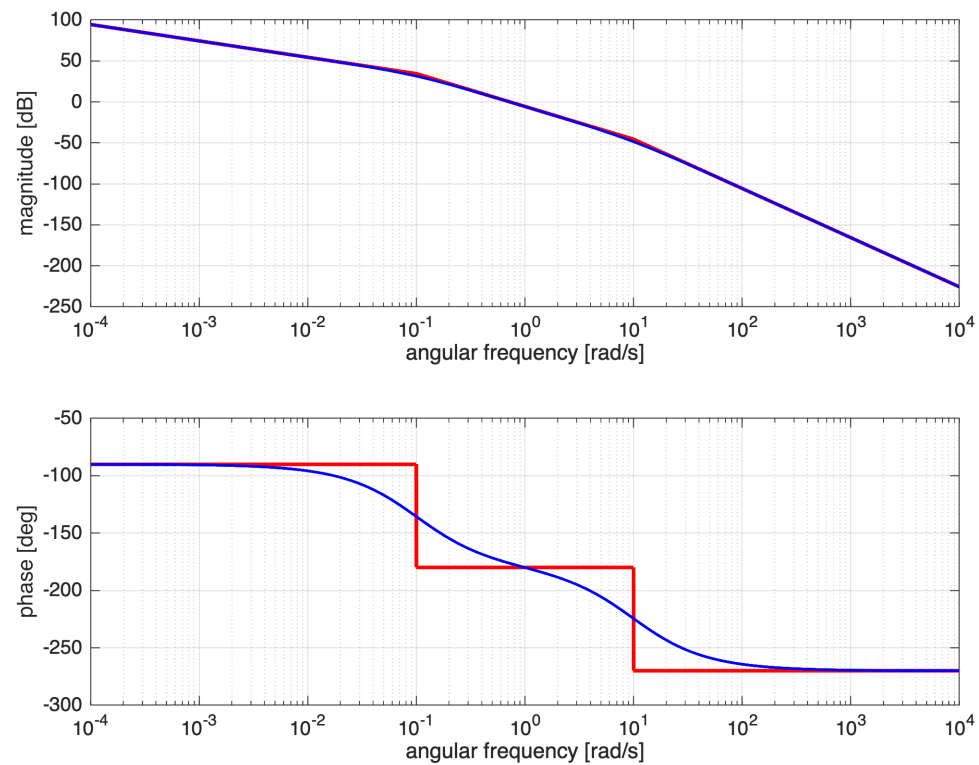
% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

% let's define the transfer function builder element
s=tf('s');

Gs = 4.815/(s*(1+s/10)*(1+10*s)); % the process G(s)

C_1s = 1.1; % a simple P controller that meets Req. (Q1.1)
Ls = C_1s*Gs;

figure;
drawBodediagrams(Ls)
```



Remark: as you can notice, the phase margin assumes values such that $\varphi_m > 75^\circ$ if and only if $\omega_c \leq 0.01$ rad/s. We are definitively unable to meet all the requirements.

Loop Shaping: "Canceling" the Slow Pole

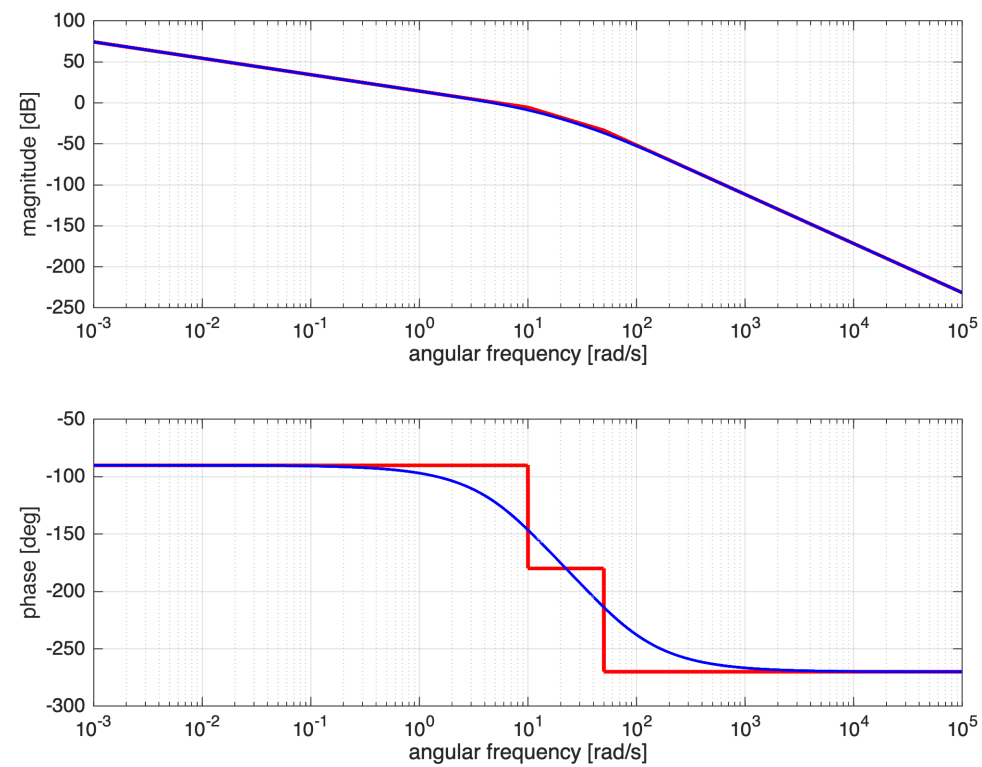
```

C2_s = (1+10*s)/(1+s/50); % let's cancel the "slow" pole
                        % and add a "fast" one
Cs = C_1s * C2_s;

Ls_trial1 = minreal(Cs*Gs);

figure;
drawBodediagrams(Ls_trial1)

```



```

rmpath(genpath('BodeDiagram/'))

```

A **visual inspection** of the phase diagram shows that at the angular frequency $\bar{\omega} = 5$ rad/s the phase of the frequency response is definitely greater than -135° , so that requirement Q1.3 is certainly fulfilled.

```

bar_omega = 5; % rad/s -- What if it was the crossover frequency?

```

Let's compute what could be the phase margin if $\omega_c = \bar{\omega}$:

```

argLbarOM = rad2deg(angle(freqresp(Ls_trial1, bar_omega)))

```

```

argLbarOM =
-122.2756

```

```

((argLbarOM+180)>45) % Req. (Q1.3)

```

```

ans = logical
1

```

Requirement Q1.3 would be fulfilled. What about requirement Q1.2?

```
((argLbar0M+180)*bar_omega) > 230)

ans = logical
      1
```

Req. (Q1.2) would also be fulfilled.

What about Req. (Q1.1)? Let's compute the gain μ_1 in order to impose $\omega_c = \bar{\omega}$

$$\mu_1 : |L(j\bar{\omega})| = 1 \implies \left| \frac{4.815 \mu_1}{j\bar{\omega} \left(1 + j\frac{\bar{\omega}}{10}\right) \left(1 + j\frac{\bar{\omega}}{50}\right)} \right| = 1 \implies \frac{4.815 \mu_1}{5 \sqrt{1 + \frac{1}{4}} \sqrt{1 + \frac{1}{100}}} = 1$$

Thus

$$\mu_1 = \frac{5 \sqrt{\frac{5}{4}} \sqrt{\frac{101}{100}}}{4.815} = \frac{\sqrt{505}}{4 \cdot 4.815} \approx 1.167 \implies \mu_1 > \bar{\mu}_1$$

Requirement Q1.1 would be fulfilled.

Summing Up:

Using the controller

$$C(s) = 1.167 \cdot \frac{1 + 10s}{1 + \frac{s}{50}}$$

all the requirements are met.

Fine Tuning

Let's import the controller in the Control System Designer app for a fine tuning.

```
Cs = 1.167*(1+10*s)/(1+s/50);

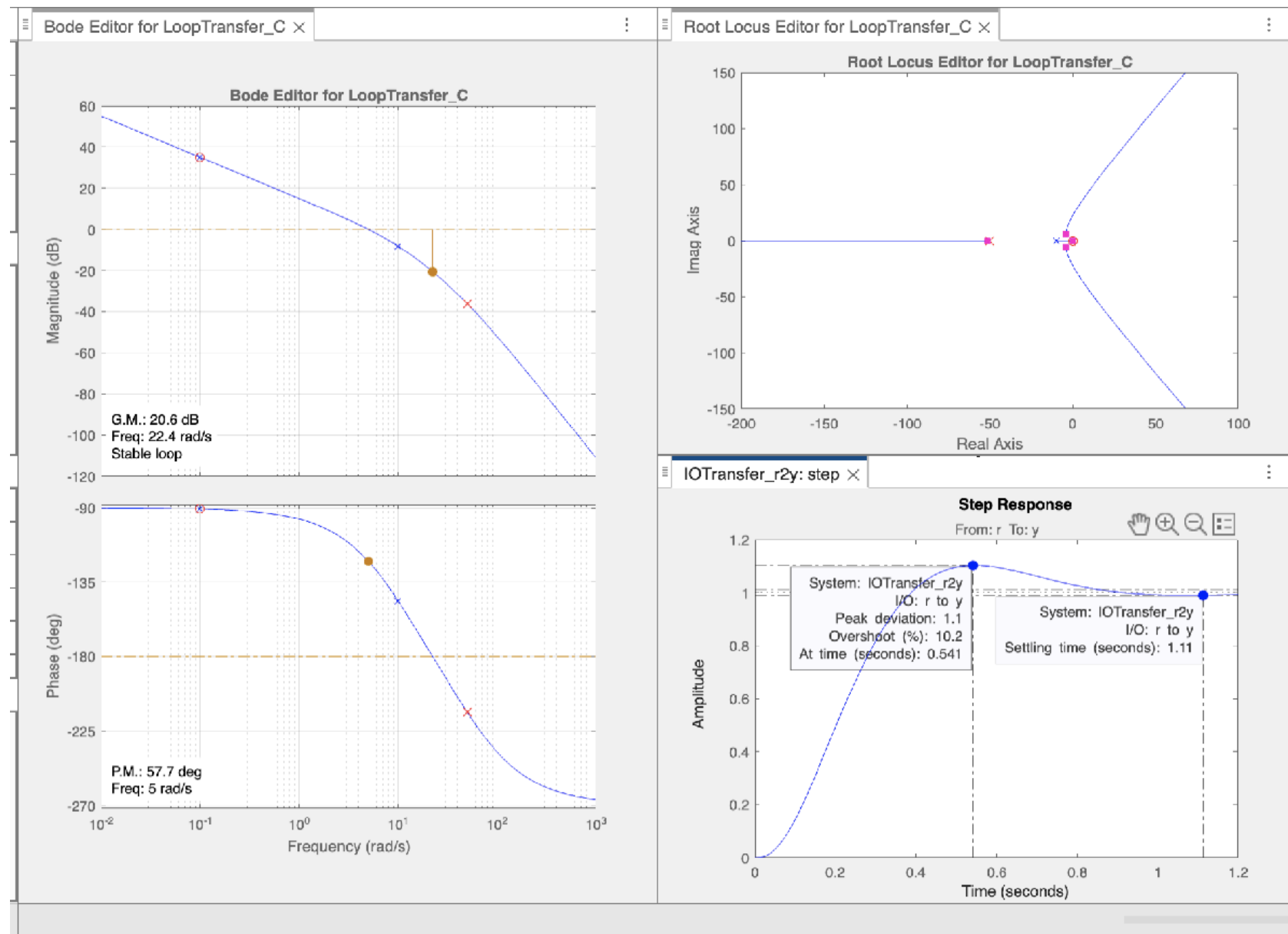
Ls = minreal(Cs*Gs)

Ls =

      2810
-----
s^3 + 60 s^2 + 500 s

Continuous-time transfer function.
Model Properties
```

controlSystemDesigner(Gs, Cs)



No needs to more fine tuning: all the requirements are met. The session MAT-file corresponding to this Figure is CSD_Solution1.mat .

Solution 2

Design Strategy

- Cancel both the poles in $p_2 = -0.1$ and $p_3 = -10$ and insert two other "very fast" poles, with the aim to enlarge the interval of angular frequencies corresponding to the phase margin $\hat{\varphi}_m \approx +90^\circ$

Design Procedure

Let's select the static controller $C_1(s) = \mu_1$, with $\mu_1 > \bar{\mu}_1$ in order to fulfill the req. (Q1.1). Moreover, select as dynamic controller $C_2(s)$ the following

$$C_2(s) = \frac{(1 + 10s) \left(1 + \frac{s}{10}\right)}{\left(1 + \frac{s}{150}\right) \left(1 + \frac{s}{175}\right)}$$

```
clear

% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

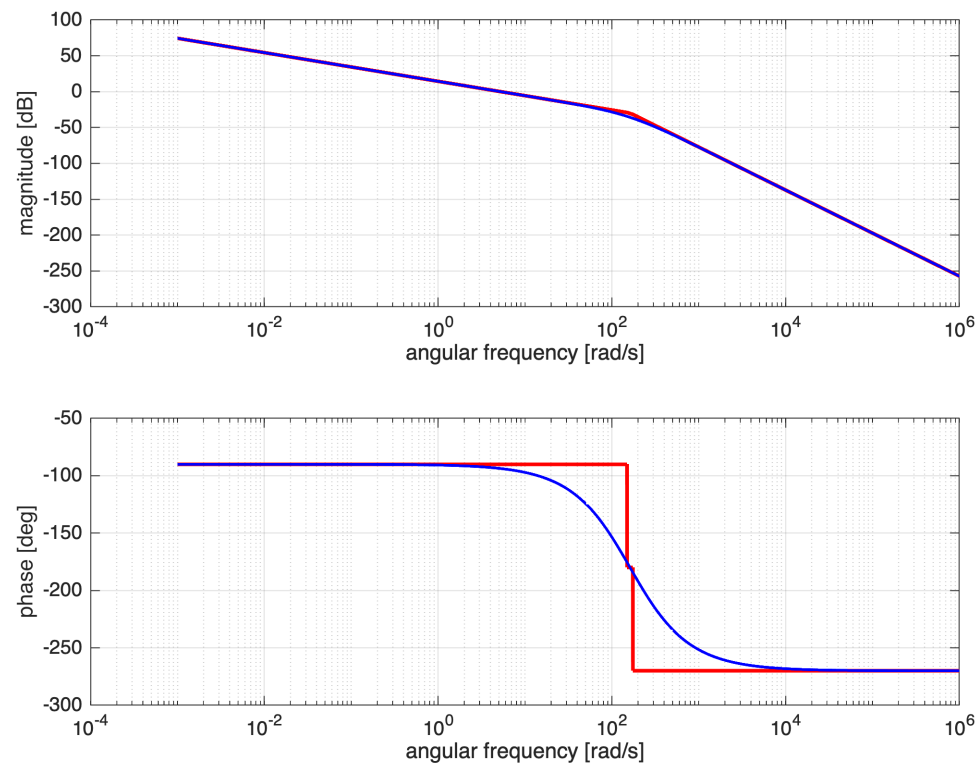
% let's define the transfer function builder element
s=tf('s');

Gs = 4.815/(s*(1+s/10)*(1+10*s)); % the process G(s)

C_1s = 1.1; % a simple P controller that meets Req. (Q1.1)
C_2s = (1+10*s)*(1+s/10)/((1+s/150)*(1+s/175));
Cs = C_1s*C_2s;

Ls = minreal(Cs*Gs);

figure;
drawBodediagrams(Ls)
```



```
rmpath(genpath('BodeDiagram/'))
```

A **visual inspection** of the phase diagram shows that at the angular frequency $\bar{\omega} = 10$ rad/s the phase of the frequency response is approximately -90° , so that **both requirements (Q1.2) and (Q1.3) are certainly fulfilled**.

```
bar_omega = 10; % rad/s -- What if it was the crossover frequency?
```

Let's compute what could be the phase margin if $\omega_c = \bar{\omega}$:

```
argLbar0M = rad2deg(angle(freqresp(Ls, bar_omega)))

argLbar0M =
-97.0846

((argLbar0M+180)>75) & (bar_omega > 2.3) % Reqs (Q1.3) & (Q1.2) both are met.

ans = logical
     1
```

What about Req. (Q1.1)? Let's compute the gain μ_1 in order to impose $\omega_c = \bar{\omega}$

$$\mu_1 : |L(j\bar{\omega})| = 1 \implies \left| \frac{4.815 \mu_1}{j\bar{\omega} \left(1 + j \frac{\bar{\omega}}{150}\right) \left(1 + j \frac{\bar{\omega}}{175}\right)} \right| = 1 \implies \frac{4.815 \mu_1}{10 \sqrt{\frac{226}{225}} \sqrt{\frac{1229}{1225}}} = 1$$

Thus

$$\mu_1 = \frac{10 \sqrt{\frac{226}{225}} \sqrt{\frac{1229}{1225}}}{4.815} \approx 2.085 \implies \mu_1 > \bar{\mu}_1$$

Requirement Q1.1 would be fulfilled.

Summing Up:

Using the controller

$$C(s) = 2.085 \cdot \frac{(1 + 10s) \left(1 + \frac{s}{10}\right)}{\left(1 + \frac{s}{150}\right) \left(1 + \frac{s}{175}\right)}$$

all the requirements are met.

Fine Tuning

Let's import the controller in the Control System Designer app for a fine tuning.

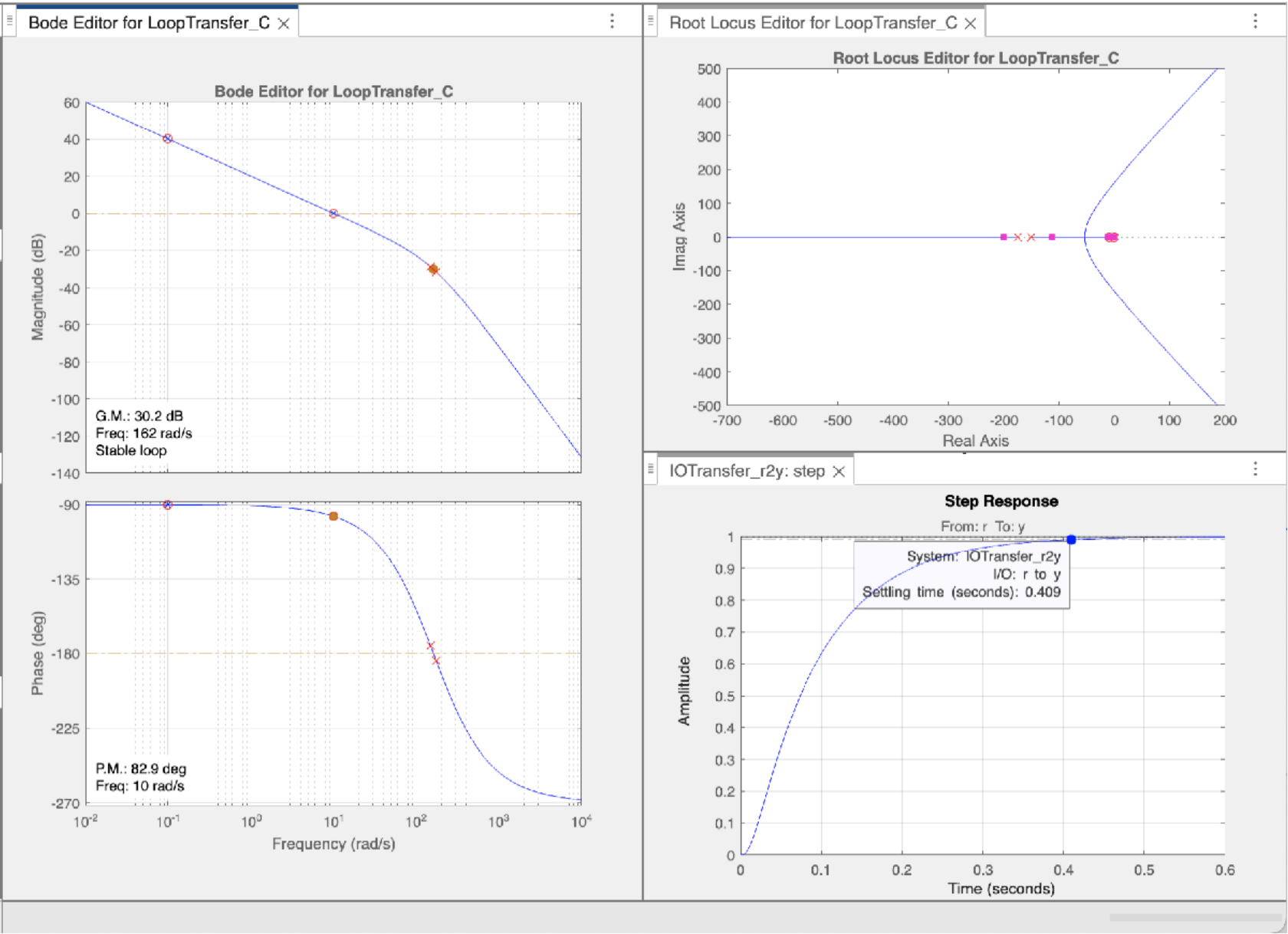
```
Cs = 2.085*(1+10*s)*(1+s/10)/((1+s/150)*(1+s/175));  
  
Ls = minreal(Cs*Gs)
```

Ls =

$$\frac{2.635e05}{s^3 + 325 s^2 + 2.625e04 s}$$

Continuous-time transfer function.
Model Properties

```
controlSystemDesigner(Gs, Cs)
```



No needs to more fine tuning: all the requirements are met. The session MAT-file corresponding to this Figure is CSD_Solution2.mat .